



A Level • Cambridge (CIE) • Chemistry

54 mins 29 questions

Multiple Choice Practice

Topical Practice MCQ

14. Hydrocarbons

Carefully selected past paper questions by topic.

27 But-2-ene reacts with cold dilute acidified KMnO_4 to give product X.

But-2-ene reacts with hot concentrated acidified KMnO_4 to give product Y.

Which statement about product X and product Y is correct?

- A** Both product X and product Y will react with 2,4-dinitrophenylhydrazine.
- B** Neither product X nor product Y will react with 2,4-dinitrophenylhydrazine.
- C** Product X will react with 2,4-dinitrophenylhydrazine, product Y will not.
- D** Product Y will react with 2,4-dinitrophenylhydrazine, product X will not.

26 Z is a gaseous hydrocarbon which has a density of $3.50 \times 10^{-3} \text{ g cm}^{-3}$ under room conditions.

Z reacts with an excess of hot concentrated acidified KMnO_4 . Only **one** type of carboxylic acid is formed in this reaction.

What is Z?

- A** but-2-ene
- B** 2,3-dimethylbut-2-ene
- C** hex-2-ene
- D** hex-3-ene

29 But-2-ene reacts with cold dilute acidified KMnO_4 to give product X.

But-2-ene reacts with an excess of hot concentrated acidified KMnO_4 to give product Y.

Which statement about X and Y is correct?

- A** Only one of X and Y reacts with 2,4-dinitrophenylhydrazine.
- B** X and Y both react with sodium hydroxide.
- C** X and Y both react with sodium metal.
- D** Y reacts with LiAlH_4 to give X.

28 Structural isomerism and stereoisomerism should be considered when answering this question.

A set of isomeric hydrocarbons:

- all contain 14.3% by mass of hydrogen
- all react with bromine by addition, 0.280 g of each hydrocarbon reacting with 0.799 g of bromine.

What is the maximum number of isomeric compounds in the set?

- A** 1 **B** 3 **C** 4 **D** 5

- 28** Ethane reacts with an excess of chlorine in the presence of ultraviolet light to form a mixture of products.

How many of these products contain two carbon atoms and one or more chlorine atoms?

A 6

B 7

C 8

D 9

31 A mixture of ethane and an excess of chlorine is exposed to UV light.

How many different products, each containing only two carbon atoms and at least one chlorine atom per molecule, can be formed?

A 8

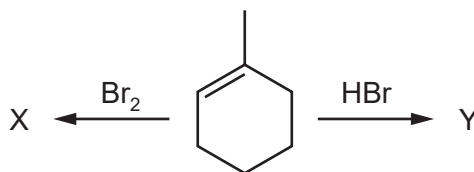
B 9

C 10

D 12

32 When 1-methylcyclohexene reacts with Br_2 the product is X.

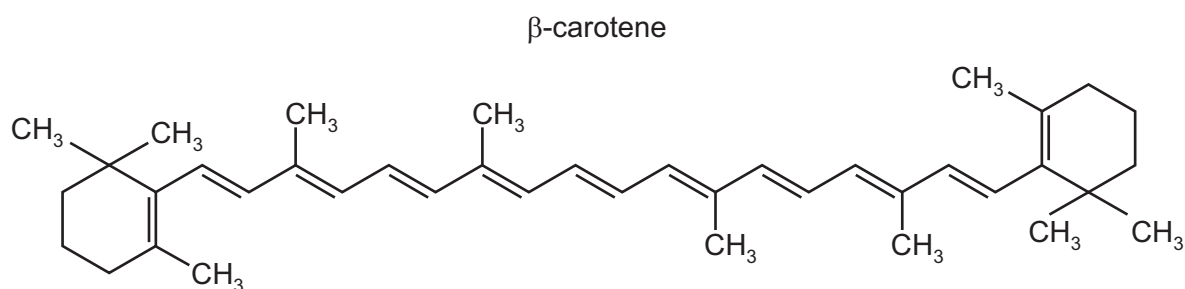
When 1-methylcyclohexene reacts with HBr the **major** product is Y.



Which statement is correct?

- A X is a mixture of two stereoisomers; Y does **not** have stereoisomers.
- B X is a mixture of two stereoisomers; Y is a mixture of four stereoisomers.
- C X is a mixture of four stereoisomers; Y does **not** have stereoisomers.
- D X is a mixture of four stereoisomers; Y is a mixture of four stereoisomers.

- 30 β -carotene is responsible for the orange colour of carrots.



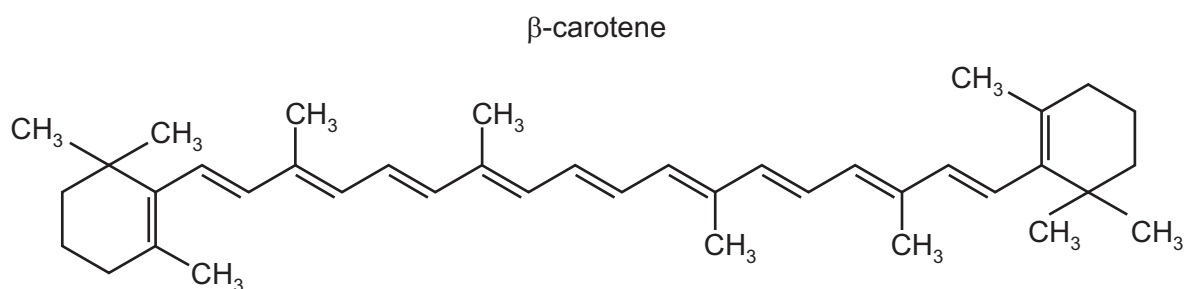
β -carotene is oxidised by hot, concentrated, acidified KMnO_4 .

When an individual molecule of β -carotene is oxidised in this way, many product molecules are formed.

How many of these product molecules contain a ketone functional group?

- A** 4 **B** 6 **C** 9 **D** 11

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- A** 4 **B** 6 **C** 9 **D** 11

30 Oct-1-ene, $\text{CH}_3(\text{CH}_2)_5\text{CH}=\text{CH}_2$, can be thermally cracked.

Which of the compounds W, X, Y and Z can be obtained by thermally cracking oct-1-ene?

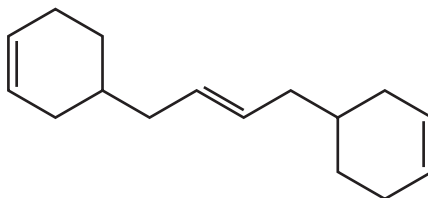
W	X	Y	Z
$\text{CH}_2=\text{CH}_2$	$\text{CH}_3\text{CH}=\text{CH}_2$	$\text{CH}_3\text{CH}_2\text{CH}_3$	$\text{CH}_2=\text{CHCH}=\text{CH}_2$

- A W, X, Y and Z
- B W, X and Y only
- C W, X and Z only
- D W and X only

21 Structural isomerism **only** should be considered when answering this question.

Molecule X contains three C=C double bonds. One mole of X is reacted with three moles of HBr. The carbon skeleton is unchanged.

molecule X



How many different products are formed?

A 3

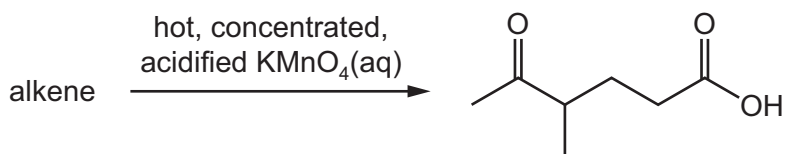
B 4

C 6

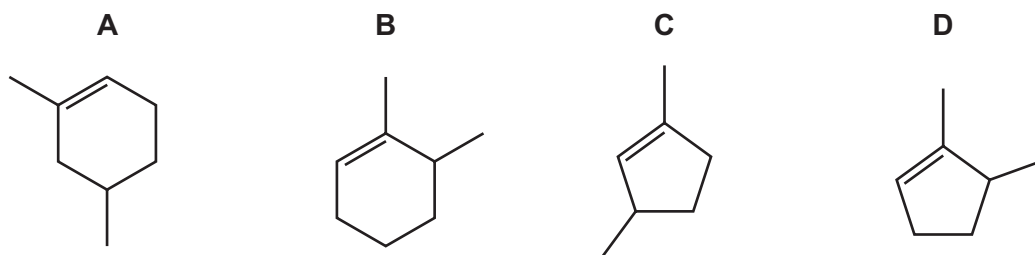
D 8

- 28** Which statement about the use of alkane fuels in internal combustion engines is correct?
- A** C_8H_{18} is used as fuel in internal combustion engines and reacts with oxygen and nitrogen from the air.
 - B** In limited oxygen, CO is produced which oxidises SO_2 to SO_3 in the atmosphere.
 - C** The catalytic converter removes polluting gases including NO_2 and CO_2 .
 - D** Unburnt hydrocarbons and NO_2 can react in sunlight to produce photochemical smog.

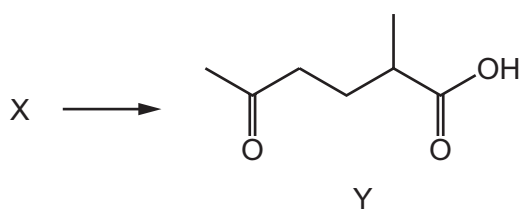
- 22 An alkene reacts with hot, concentrated, acidified potassium manganate(VII) to produce a single organic product as shown.



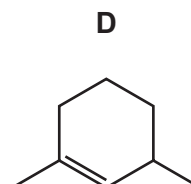
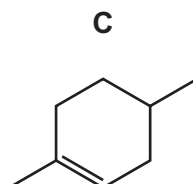
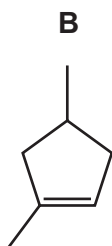
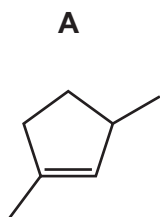
What is the structure of the alkene?



22 Compound X can be converted into compound Y in a single step.



What could be the identity of X?

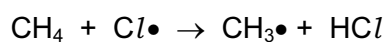


- 24** 2,3-dimethylpent-2-ene, $(\text{CH}_3)_2\text{C}=\text{C}(\text{CH}_3)\text{CH}_2\text{CH}_3$, is treated with cold, dilute KMnO_4 . The product of this reaction is treated with an excess of concentrated H_2SO_4 at 180°C , giving a mixture of isomeric hydrocarbons with molecular formula C_7H_{12} .

What is the name of one of the isomeric hydrocarbons?

- A** 2,3-dimethylpenta-1,2-diene
- B** cis-2,3-dimethylpenta-1,3-diene
- C** 2,3-dimethylpenta-1,4-diene
- D** 3,4-dimethylpenta-1,3-diene

23 The **first** propagation step in the reaction between methane and chlorine is shown.



How many different **first** propagation steps are possible in the reaction between pentane and chlorine?

A 2

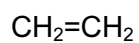
B 3

C 4

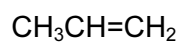
D 5

22 Oct-1-ene, $\text{CH}_3(\text{CH}_2)_5\text{CH}=\text{CH}_2$, can be thermally cracked.

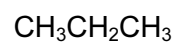
Which combination of compounds W, X, Y and Z can be obtained by thermally cracking oct-1-ene?



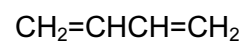
W



X



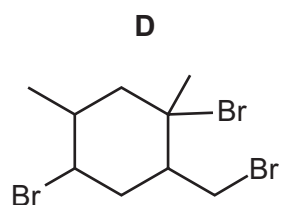
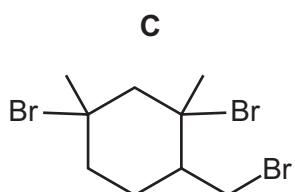
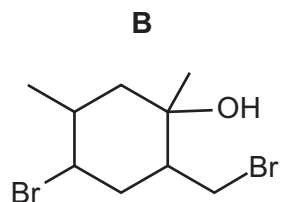
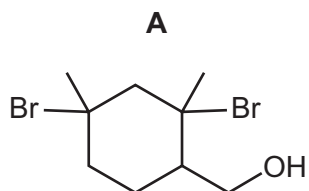
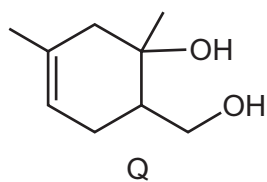
Y



Z

- A W, X, Y and Z
- B W, X and Y only
- C W, X and Z only
- D W and X only

21 What is the major product formed when compound Q is warmed with excess HBr?



26 Alkane X has molecular formula C_4H_{10} .

X reacts with $Cl_2(g)$ in the presence of sunlight to produce only two different monochloroalkanes, C_4H_9Cl . Both of these monochloroalkanes produce the same alkene Y, and no other organic products, when they are treated with hot ethanolic KOH.

What is produced when Y is treated with hot concentrated acidified $KMnO_4$?

- A CO_2 and $CH_3CH_2CO_2H$
- B CO_2 and CH_3COCH_3
- C HCO_2H and CH_3COCH_3
- D CH_3CO_2H only

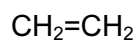
21 In the presence of ultraviolet light, ethane and chlorine react to give a mixture of products.

Which compound could be present in the mixture of products?

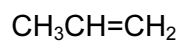
- A CH_3Cl
- B $\text{CH}_3\text{CH}_2\text{CH}_2\text{Cl}$
- C $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3$
- D $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$

24 Oct-1-ene, $\text{CH}_3(\text{CH}_2)_5\text{CH}=\text{CH}_2$, is subjected to thermal cracking.

Which combination of compounds W, X, Y and Z can be obtained?



W



X



Y



Z

- A** W, X, Y and Z
- B** W, X and Y only
- C** W, X and Z only
- D** W and X only

- 23** The cracking of a single hydrocarbon molecule, C_nH_{2n+2} , produces two hydrocarbon molecules only. Each hydrocarbon product contains the same number of carbon atoms in one molecule. Each hydrocarbon product has non-cyclic structural isomers.

What is the value of n ?

- A** 4 **B** 6 **C** 8 **D** 9

- 23** The cracking of a single hydrocarbon molecule, C_nH_{2n+2} , produces two hydrocarbon molecules only. Each hydrocarbon product contains the same number of carbon atoms in one molecule. Each hydrocarbon product has non-cyclic structural isomers.

What is the value of n ?

A 4

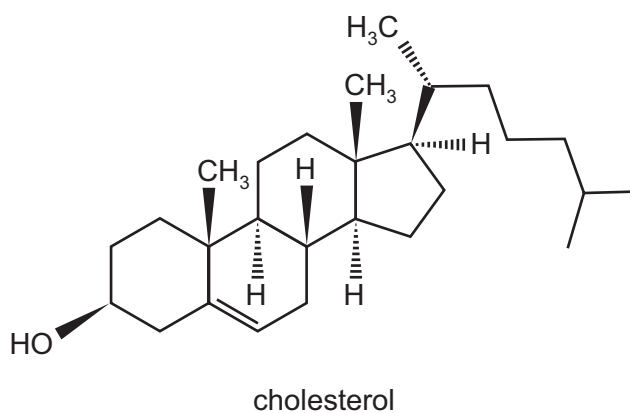
B 6

C 8

D 9

- 29 This question should be answered by considering the reactions of KMnO_4 with different functional groups under the stated conditions.

The diagram shows the structure of the naturally-occurring molecule cholesterol.



Cholesterol is separately treated with

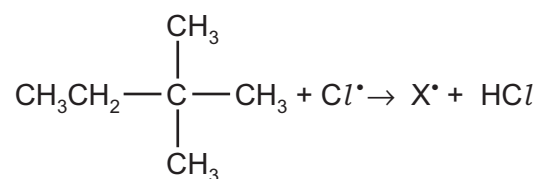
- cold, dilute acidified KMnO_4 ,
- hot, concentrated acidified KMnO_4 .

What is the change in the **number** of chiral carbon atoms in the molecule during each reaction?

	cold, dilute acidified KMnO_4	hot, concentrated acidified KMnO_4
A	+1	0
B	+1	-1
C	+2	0
D	+2	-1

- 21 When heated with chlorine, the hydrocarbon 2,2-dimethylbutane undergoes free radical substitution.

In a propagation step the free radical $X^\bullet$ is formed.

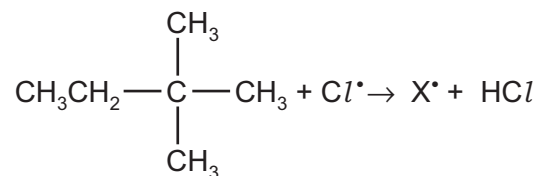


How many different forms of $X^\bullet$ are possible?

- A** 1 **B** 2 **C** 3 **D** 4

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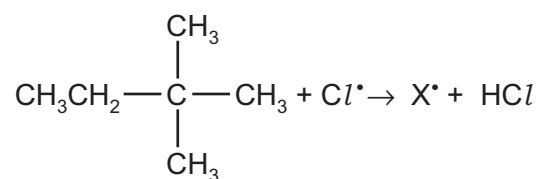


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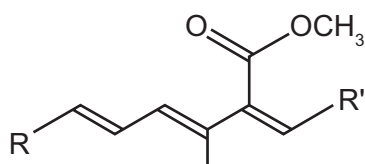
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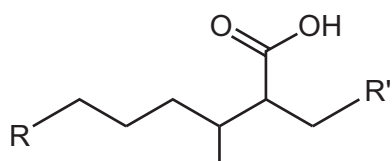
- A** 1 **B** 2 **C** 3 **D** 4

27 Part of the structure of strobilurin, a fungicide, is shown. R and R' are inert groups.

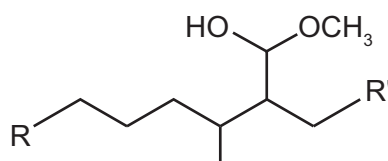


If strobilurin is first warmed with aqueous sulfuric acid, and its product then treated with hydrogen in the presence of a palladium catalyst, what could be the structure of the final product?

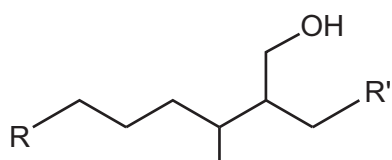
A



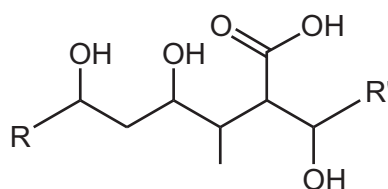
B



C



D



18 Most modern cars are fitted with three-way catalytic converters in the exhaust system.

Which three gases are removed by such a catalytic converter?

- A** carbon monoxide, hydrocarbons, nitrogen oxides
- B** carbon monoxide, carbon dioxide, nitrogen oxides
- C** carbon monoxide, nitrogen oxides, sulfur dioxide
- D** hydrocarbons, nitrogen oxides, sulfur dioxide

Answer Key

No.	Paper	Topic	Answer
1	9701_m24_qp_12	14. Hydrocarbons	B
2	9701_s24_qp_12	14. Hydrocarbons	D
3	9701_s24_qp_13	14. Hydrocarbons	C
4	9701_s23_qp_11	14. Hydrocarbons	C
5	9701_s23_qp_12	14. Hydrocarbons	D
6	9701_s23_qp_13	14. Hydrocarbons	B
7	9701_s23_qp_13	14. Hydrocarbons	C
8	9701_w23_qp_11	14. Hydrocarbons	B
9	9701_w23_qp_13	14. Hydrocarbons	B
10	9701_s22_qp_11	14. Hydrocarbons	A
11	9701_w21_qp_13	14. Hydrocarbons	B
12	9701_w20_qp_12	14. Hydrocarbons	D
13	9701_w19_qp_11	14. Hydrocarbons	D
14	9701_m18_qp_12	14. Hydrocarbons	A
15	9701_w17_qp_12	14. Hydrocarbons	B
16	9701_s16_qp_11	14. Hydrocarbons	B
17	9701_s16_qp_12	14. Hydrocarbons	A
18	9701_s15_qp_12	14. Hydrocarbons	C
19	9701_s15_qp_13	14. Hydrocarbons	B
20	9701_s14_qp_13	14. Hydrocarbons	C
21	9701_s12_qp_12	14. Hydrocarbons	A
22	9701_w12_qp_11	14. Hydrocarbons	C
23	9701_w12_qp_12	14. Hydrocarbons	C
24	9701_s11_qp_12	14. Hydrocarbons	D
25	9701_s10_qp_11	14. Hydrocarbons	C
26	9701_s10_qp_12	14. Hydrocarbons	C
27	9701_s10_qp_13	14. Hydrocarbons	C
28	9701_w10_qp_12	14. Hydrocarbons	A

Answer Key

No.	Paper	Topic	Answer
29	9701_w09_qp_11	14. Hydrocarbons	A